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Mapping and Detection of Evolving Trends in DDoS Attack Patterns using AI Driven Models

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***Abstract*—Distributed Denial-of-Service (DDoS) attacks have become a serious and emerging threat to network security. These attacks have been growing rapidly in recent times, creating a significant threat to online services and severely impacting the security of networks and users. In this study, multiple instances of different DDoS attack patterns are analyzed across various time periods to map the evolving trends in attack patterns. In the present study, a dataset is prepared, which contains benign traffic data as well as malicious ones in a specific ratio so as to simulate real world traffic and prevent over-fitting in our AI model. Our machine learning model is trained on a dataset comprising DDoS-specific entries from the CIC IoT-DIAD 2024, CIC IoT 2023, and CSE-CIC-IDS2018 datasets. A comparative evaluation is performed among various models, including Linear Regression, Random Forest Classifier and Extreme Gradient Boosting. Key figures of merits such as detection efficiency and computational efficiency are evaluated using the models. Among these, highest accuracy of 99.56% is achieved using Extreme Gradient Boosting model. Further, the model exhibits excellent results for other key performance metrics also.**

Keywords—DDoS attack, Machine Learning, Network security, Linear Regression, Extreme Gradient Boosting, Random Forest Classifier.

I. INTRODUCTION

In a DDoS attack, a server is flooded with packets, keeping it occupied with processing multiple requests and consequently degrading its performance. DDoS attacks are a critical concern due to their ability to overwhelm and incapacitate targeted systems through distributed networks. As these attacks grow in scale and sophistication, traditional rule-based detection mechanisms have proven inadequate in identifying complex and evolving threat patterns. According to Nexusguard 2025 DDoS trends report [1], HTTPS Flood is the most widely used attack type. Recent trends suggest that DNS Attacks, IP Fragmentation attacks and UDP Fragmentation attacks have also risen in popularity with respect to the previous year's data, with DNS attacks and IP Fragmentation attacks exhibiting more than 500% increase in difference by percentage. These details reinforce our need to study various DDoS attack patterns and their cyclical trends to counter them effectively.

Consequently, various machine learning (ML) models have unfolded as efficient techniques for automating and enhancing DDoS detection by learning from traffic behaviour and adapting new attack vectors. Over the last decade, several research efforts have been put on ML-based Intrusion Detection System (IDS) with high accuracy [2-26]. Some previous works in this field include proposal of a feature selection mechanism making use of the results obtained from the calculation using the CIC-IDS2017 dataset [21-23]. Usha *et al*. utilized multiple machine learning methods like *K*-Nearest Neighbor (KNN), XGBoost and CNN deep learning architecture to detect and organize DDoS attacks using the CIC-DDoS2019 dataset [24]. The main disadvantage of their approach was its high computational complexity and data imbalance. Alimi *et al*. proposed an IDS based on LSTM (Large Short Term Memory) for the detection of DoS attacks. The framework was evaluated using the CICIDS-2017 datasets [18]. The obtained results exhibited effectiveness of the LSTM, detecting DoS attacks with accuracies between 98% and 99%. Akgun *et al*. used CIC-DDoS2019 dataset and reported an algorithm for detection of these attacks [19]. The results of their work showed CNN model as best among all the proposed models. Very high accuracy of greater than 99% was obtained for both binary and multi-class classifications. However, some features with uneven distribution were used, which led to lack of generalization for effective use of this model. Many researchers explored the possible threats for the ML-based IDS [27-31]. Researchers reported better detection accuracy and reduction in the dimensionality of features considering various models such as Support Vector Machine, Quantum-Inspired Redundant and Irrelevant feature Minimization and Extraction methods. Random Forest in combination with principal component analysis was also used in improving processing speed and reduction of model complexity [32-37]. However, most of the previous works done in this field utilize outdated datasets which do not completely represent the present level of DDoS attacks sophistication. In the datasets used, the attack types included often did not target all the levels of the OSI network layer. Many outdated datasets suffered from class imbalance and other factors which contribute to prediction biasness. This included ratio of benign and malicious entries which did not resemble actual traffic patterns. Large number of entries of a single type of DDoS attack populates the dataset while other patterns were not given recognition etc. Therefore, in the present study, a dataset is prepared, which contains benign traffic data as well as malicious ones in a specific ratio so as to simulate real world traffic and prevent over-fitting in our AI model.

II. PREPARATION OF DATASET

A flow chart of the program used in the present work is presented in Fig.1. The dataset used in the present study

contains benign traffic data as well as malicious ones in the ratio of 7:3. Since real world traffic is highly variable and imbalanced, the 7:3 ratio of benign to malicious entries was taken in accordance with previous research done in this field [38]. This ratio ensures enough malicious entries are present for the ML model to detect patterns while preventing over-fitting at the same time.

The DDoS related traffic entries have been selected from CSE-CIC-IDS2018, CIC IoT 2023 and CIC IoT-DIAD 2024. Given the diversity of the databases used, utmost care has been taken to ensure that the attributes present in each individual database has been there in the final custom database.

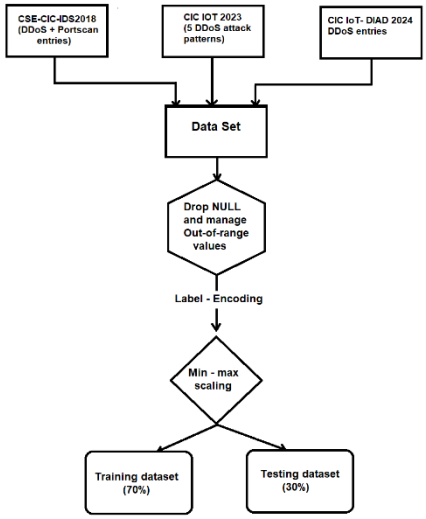


TABLE I
THE DISTRIBUTION OF VARIOUS ATTACK VARIANTS AND THEIR TARGET NETWORK OSI LAYER IN OUR PREPARED DATASET

Attack Type	Target OSI Layer	Selected Entries
DDoS-SlowLoris	Application	2,343
DDoS-HTTP Flood	Application	2,877
HOIC + LOIC	Application/ Transport	1,28,027
DNS Spoofing	Application	17,897
DDoS-SYNFlood	Transport	59,811
DDoS-RSTFINFlood	Transport	30,000
DDoS-PSHACK Flood	Transport	35,000
DDoS-TCP Flood	Transport	25,000
DDoS-ICMP Flood	Network	90,000
DDoS- ICMP Fragmentation	Network	35,000
DDoS-UDP Fragmentation	Network	28,683
DDoS-SynonymousIPFlood	Network	86,993
Benign	-	16,72,837
TOTAL	-	22,14,468

TABLE I shows the distribution of various attack variants and their target Network OSI layer in our prepared dataset. This contains popular and most frequently used attack types in a ratio which gives proper representation to each network OSI layer. Here, different databases have different column names for the same feature, those columns have not been repeated and a single column header has been fixed. The range of the time stamp of the DDoS attack vectors has been from 2018 to handle common attack methods and populate the dataset with entries targeting the application layer and extend till 2024 dataset. This has encompassed one of the newest attack patterns in today's world. These attack patterns have included SlowLoris, ACK fragmentation, ICMP flood, UDP flood, HTTP flood, UDP fragmentation, TCP flood, SYN flood etc. The data has been extracted using CIC Flow meter and contain over 80 attributes. The common attributes within the three datasets have been selected and all the basic pre-processing tasks have been performed. The pre-processing has included removal of null values, managing out-of-range values and label encoding the dataset so that it contains values of the data type suitable for the models. After the pre-processing was completed, the dataset was split into train and test sets with equal ratio of malicious and benign entries in both sets, using the 'stratify' parameter. This was done because the population of some attack types was more than the others, and the less populated attacks also require equal representation.

From these, 15 best attributes have been selected using scikit-learn's (sklearn) SelectKBest library. SelectKBest [Eq.(1)] in sklearn has been used to choose the best k features from a dataset based on a scoring function.

$$F = \frac{\frac{1}{N_k}\sum_{i=1}^{N_k}(\overline{X}_k - \overline{X})^2}{\frac{1}{N}\sum_{i=1}^{N}(X_i - \overline{X})^2} \quad (1)$$

Where, N_k is the feature value for each class, $\overline{X}_k$ is the mean of class k feature values, $\overline{X}$is the overall average, N_k denotes total samples in class k and N refers to total number of samples. It was based on mutual_info_classif (mutual information classifier). With the present classif (classifier) as the scoring function, it selects the features that have the highest mutual information with the target variable. This can help identifying the most relevant features for a classification problem. Its mathematical formula is given by Eq. (2).

$$\mathrm{I}(X,Y) = \sum_{z\in X}\sum_{y=Y} p\,(x,y) log\left(\frac{p(x,y)}{p(x)p(y)}\right) \quad (2)$$

Where, p(x,y) denotes combined probability distribution of X and Y and $p(x)$, $p(y)$ are individual probabilities of X and Y, respectively. SelectKbest chooses the features based purely on a mathematical relation of the attribute values with other elements of the dataset and has no prior technical knowledge of cyber security.

Despite this, it is observed that the features selected from the vast available 80 attributes are all crucial and universally applicable for all DDoS attack variations. This validates the suitability of this feature extraction technique and is confirmed by the effect of these attributes on the final model output, which will be discussed in the later section.

III. RELATION AMONG VARIOUS ATTACK PATTERNS

From the available attack patterns, 15 contrasting variants have been selected from the subset of CIC IoT-DIAD 2024

and have mapped their respective attribute values of rate, srate (outbound packet transmission rate), iat (inter arrival time), flag offsets such as syn_count and fin_count, protocol.

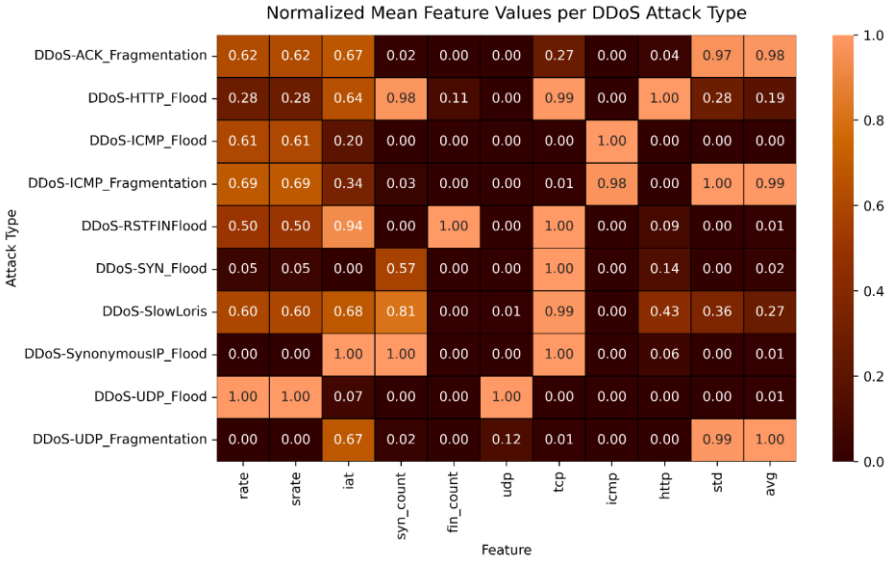


Fig.2. A map of attack type versus features

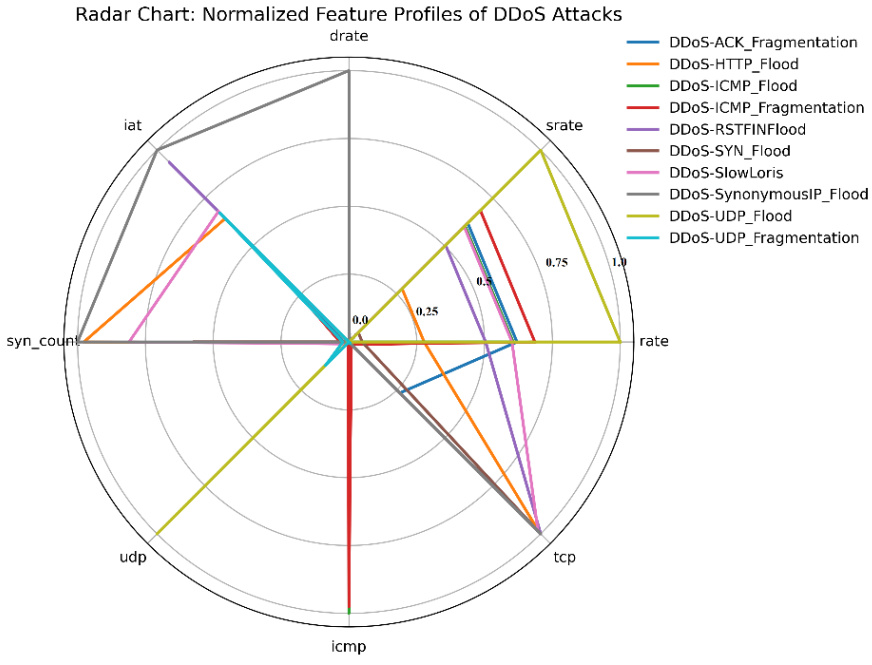


Fig.3. A radar chart of normalized feature profiles of DDoS attacks.

offsets such as UDP, TCP, ICMP, HTTP, std (standard deviation of packet length) and avg (average packet length) in the heatmap given in Fig.2. A radar chart is also depicted using normalized values of selected features to show unique dependence of certain DDoS variants on individual attributes.

A radar chart is shown in Fig.3. which signifies that for "protocol" column, "icmp" entries of the dataset have potential to identify DDoS-ICMP attack types. Similarly, "iat" column entries show high degree of relation with the DDoS-RSTFINF flood attack variant. This shows the variety and diversity of the attack vectors and confirms that the familiarity of a model with a single attack type does not guarantee detection of other DDoS variants. Therefore, our model must be trained on a dataset which contains a large variety of DDoS attack patterns.

A. Three Similar Attributes

The graphs shown in Fig.4(a), (b), and (c) map the density of values distributed across the dataset for their respective attributes. These show the attributes for which benign and DDoS traffic is similar, and which are distinct. Visual representation of the dataset can also help us identify which common attributes across all DDoS attack variants are

significantly distinct from benign traffic. The graphs are plotted using matplotlib.pyplot library of python where x axis represents column (feature) values and y axis represents density or the likeliness of the occurrence of the value.

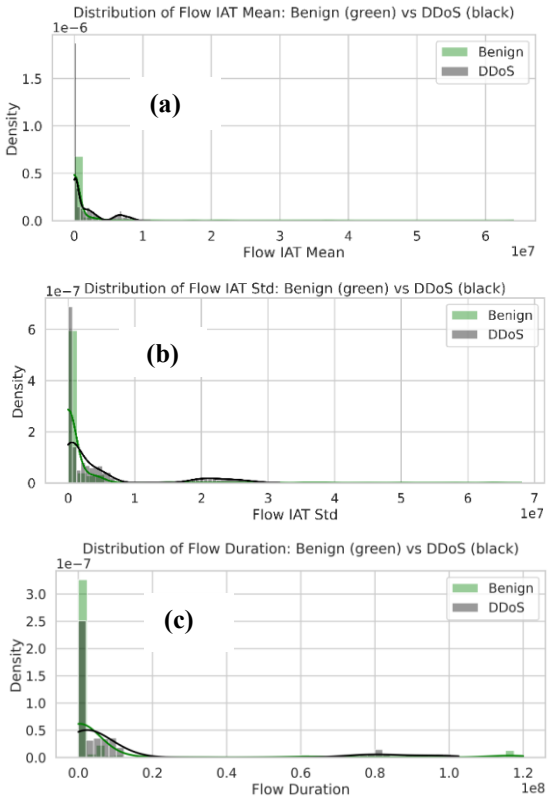


Fig.4. Mapping of density of values across the data set for their attributes (a) Distribution of Flow IAT mean. (b) Distribution of Flow IAT Std, and (c) Distribution of Flow duration.

In the obtained graphs, the range of values and units depend on the feature. For example, the unit for “Flow Duration” is micro seconds while the unit for “Max Packet Length” is bytes. Likewise, the higher the probability of occurrence of the value, the higher will be the position of the graph’s y axis. This can be cross verified with the results obtained from SelectKBest with mutual_info_classif metric. On comparing the “Flow IAT” values between malicious and benign traffic entries, less degree of distinction is observed. IAT (Inter Arrival Time) std. refers to the standard deviation of time between two flows. Flow IAT Mean refers to the mean time between two flows. This shows the ability of modern DDoS attack variants to bypass traditional IDS systems by containing attributes of Flow IAT or flow duration similar to that of benign traffic.

B. Three Different Attributes

A Distribution of “rate” across various DDoS attack types is shown in Fig.5.

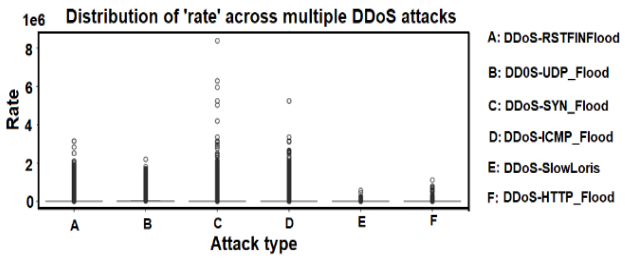


Fig.5. Distribution of “rate” across various DDoS attack types.

Earlier DoS attack variants were characterized for extremely high rates of dataflow. However, recent attack patterns have evolved, which show variable or even low rates.

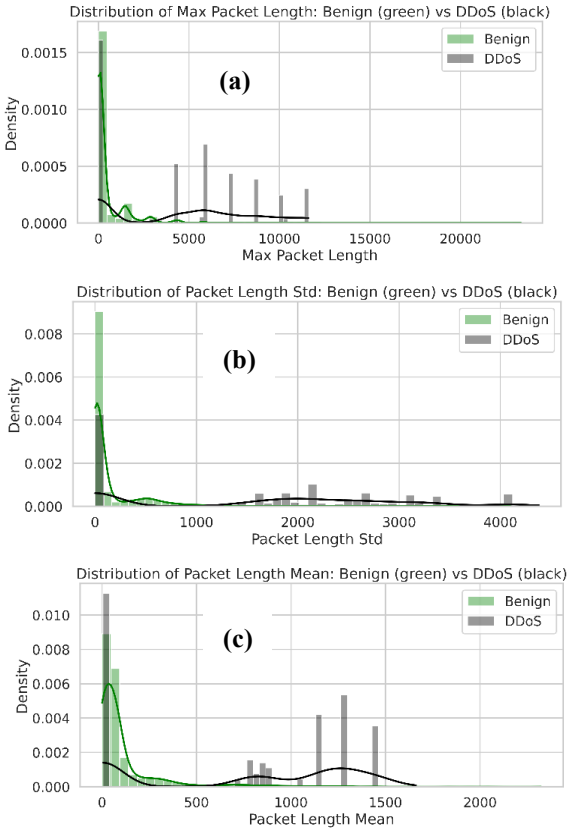


Fig.6. (a) Mapping of density of values across the data set for their attributes (a) Distribution of maximum packet length, (b) Distribution of packet length standard. (c) Distribution of packet length mean.

Other than that, various other attributes of specific DDoS attacks are similar to that of benign traffic, which makes it difficult to segregate them. Our dataset’s large variety and magnitude of entries ensure that the model has sufficient data to find links and hidden patterns to differentiate between benign and malicious traffic. Despite their characteristics of imitating benign traffic features, some attributes of malicious traffic related to packet length are clearly distinct. It shows a shift from earlier cases where Flow rate or Packets/s used to be distinct to current DDoS variants being able to show similar rate and packet speed features. But they are still unable to disguise their packet length characteristics to be similar to that of benign traffic.

C. Model Training

To improve the performance of our model, certain features were prioritized in order to reduce the dimensionality. This ensured that the model only focused on relevant attributes. The attributes are ‘Destination Port’ (DP), ‘Total Length of Forward Packets’ (TLFP), ‘Total Length of Backward Packets’ (TLBP), ‘Maximum Forward Packet Length’ (MFPL), ‘Maximum Backward Packet Length’ (MBPL), ‘Backward Packet Length Mean (BPLM), ‘Maximum Packet Length’ (MPL), ‘Packet Length Mean’ (PLM), ‘Packet

Length Std (PLS), 'Packet Length Variance' (PLV), 'Average Packet Size' (APS), 'Average Backward Segment Size' (ABSS), 'Sub-flow Forward Bytes' (SFB), 'Sub-flow Backward Bytes' (SBB) and 'Init-Win-Bytes-Forward' (IWBF). These columns were chosen using SelectKBest. However, these values are dependent on the data of attack types used. For example, the 'Protocol' flag will be helpful to classify UDP/TCP DDoS attacks but are not as useful for general classification of other DDoS attack patterns. Since this is a case of binary classification and not a multi-class one, predicting the type of attack is not as important as segregating malicious traffic from benign ones. Three models have been trained on the pre-processed dataset. Models used were Logistic Regression (LR), Random Forest Classifier (RFC), and Extreme Gradient Boosting (XGBoost).

The AI models chosen are good enough to use these selected features to successfully identify these attack patterns too. Mapping of density of values across the data set for three different attributes i.e. maximum packet length, packet length standard and packet length mean is plotted in Fig.6(a), (b), and (c).

D. SHAP Value

After the initial use of SelectKBest via mutual_info_classif metric, the actual impact of each feature on our model's output was clearly observed. Its effect has been mapped using Eq.(3). The SHAP (SHAPely value) is determined by a mathematical formula (Eq.3) which is represented below.

$$\varphi_i(f) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \qquad (3)$$

Where, $\varphi_i(f)$ is the SHAP value of feature i, N is a set containing all feature values, S is a subset of features excluding$i$, $f(S)$ is the model's prediction using the features in subset S, $f(S \cup \{i\})$ is the model's updated prediction after the inclusion of feature i into subset S, $|S|$ is the feature count in subset S and$|N|$ is the total number of features.The bee swarm graph shows the positive as well as negative impact on the predicted result (Fig.7).

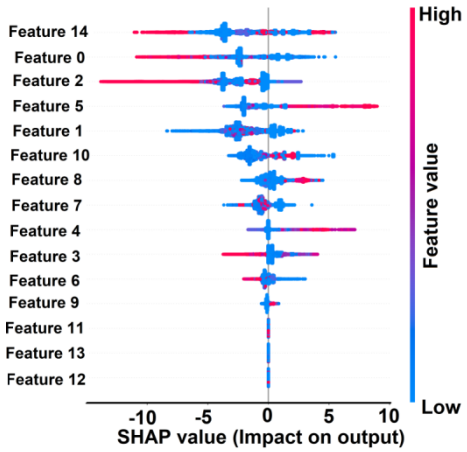


Fig.7. Plot of SHAP value vs. features.

This observation supports our decision to select certain features to be given more priority than the remaining attributes.

Reference for features- 0: DP; 1: TLFP; 2: TLBP; 3: MFPL; 4: MBPL; 5: BPLM; 6: MPL; 7: PLM; 8: PLS; 9: PLV; 10: APS; 11: ABSS; 12: SFB; 13: SBB; 14: IWBF [Fig.7].

IV. RESULTS

The experimental results obtained after training multiple models on our feature engineered dataset are given below. The evaluation metrics utilized for comparing the model output is accuracy, but other factors such as recall and f1 score are also compared for checking the overall effectiveness of the model for DDoS detection. In addition, confusion matrices depicting the true positive, false positive, true negative and false negative are also shown for better understanding of the model predictions and their drawbacks (Fig.8). LR, a linear model, provides rapid training and inference times and serves as a baseline due to its simplicity and interpretability.

TABLE II
KEY FIGURE OF MERITS OBTAINED FROM THE THREE DIFFERENT MODELS USED IN THE STUDY

Model Name	Accuracy %	Precision %	Recall %	F1-score %
XGBoost	99.56	99.30	98.92	99.11
Random Forest Classifier	99.29	98.72	98.39	98.56
Logistic Regression	87.00	79.00	66.00	72.00

However, its limited capacity to model non-linear relationships restricts its detection accuracy to 87%. The Receiver Operating Characteristic (ROC) curve for the test set of the logistic regression model is given in Fig.9(a), which shows deviation from ideal behavior. Overall performance of the model is depicted through a confusion matrix, through which it can be inferred that the model prediction has a relatively high false negative count. Random Forest, an ensemble-based method, improves accuracy by capturing feature interactions and reducing over fitting, albeit with increased training time [Fig.9(b)]. This model gives an impressive accuracy of 99.29%, which can be seen from the given confusion matrix. These results are obtained by tuning the parameters of the random Forest Classifier model and their values set as number of estimators equal to 80, maximum depth equal to 1, minimum sample leaves equal to 20 and minimum samples split equal to 50. XGBoost further enhances performance through gradient boosting, regularization and effective handling. Eq.(4) and (5) were used for calculation as given below.

$$obj(\theta) = L(\theta) + \Omega(\theta) \qquad (4)$$

Where, L denotes the loss function for training and Ω is the regularization term while θ is the best fit parameter. The loss function is generally taken to be

$$L(\theta) = \sum_i (y_i - \hat{y}_i)^2 \qquad (5)$$

The XGBoost model outperforms both LR and RFC as it is more complex and is able to find the patterns between various DDoS attack types effectively. The result is shown in Fig. 9(c).

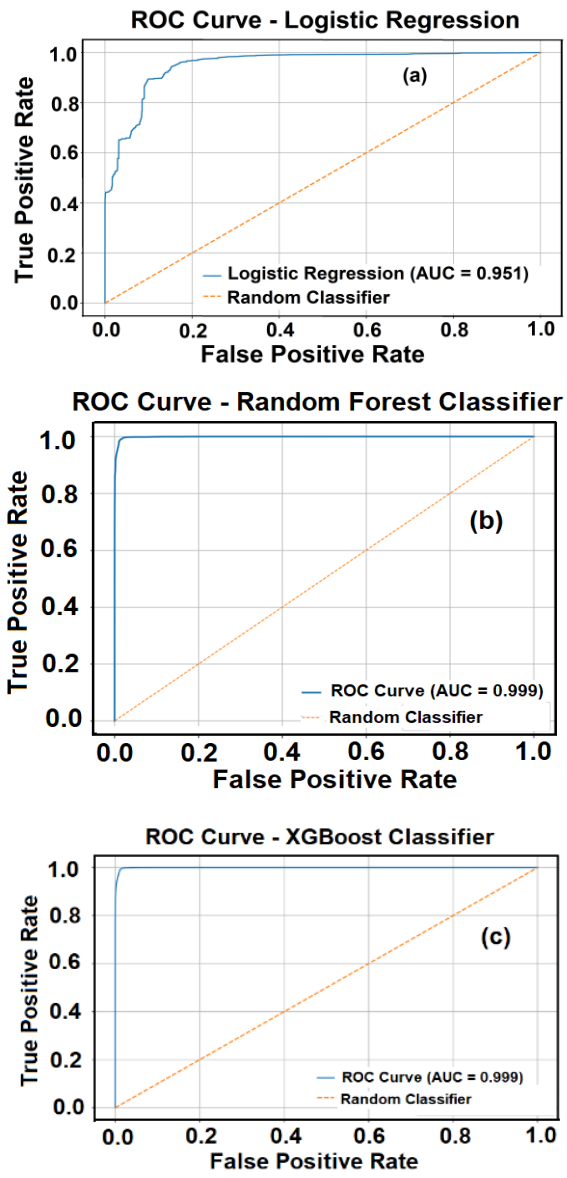


Fig.8. ROC curves plotted using (a) Logistic Regression, (b) Random Forest Classifier, (c) XGBoost. Each curve is compared with the same of Random Classifier (dashed orange line). AUC value of each curve is presented for respective model.

To validate the model's performance with an external dataset, cross-dataset testing was performed on a subset of the CIC-DDoS2019 dataset [23] with 500000 extracted entries. This dataset contained a vast amount of DNS, NTP, UDP, SSDP and SYN variants of DDoS attack data. To overcome the class imbalance, 300000 malicious labels were replaced by unused benign entries of the previous dataset into the new test set. The same scaler was fit on the model for normalizing the values and similar pre-processing tasks were performed before passing it as the test set into the model for evaluation. As a result, the XGBoost model gave an impressive accuracy of 98.9%, precision of 98.3%, recall of 99.0% and an f1 score of 98.6%. This shows that our model is capable of performing under new, external datasets.

This comparative analysis serves as a foundation for selecting and optimizing ML-based DDoS detection strategies suitable to specific operational requirements. The results obtained using three models i.e. LR, RFC and XGBoost for the key parameters are consolidated in TABLE II. Results show significant improvement upon using XGBoost model confirming the superiority of the model over others.

A comparison has been made among the results obtained in the previous reported works and in the present study, which is furnished in TABLE III. The comparison exhibits advantage of using the present dataset after feature extraction combined with XGBoost model, which manifests into significant improvement in the key figure of merits of the DDoS detection scheme.

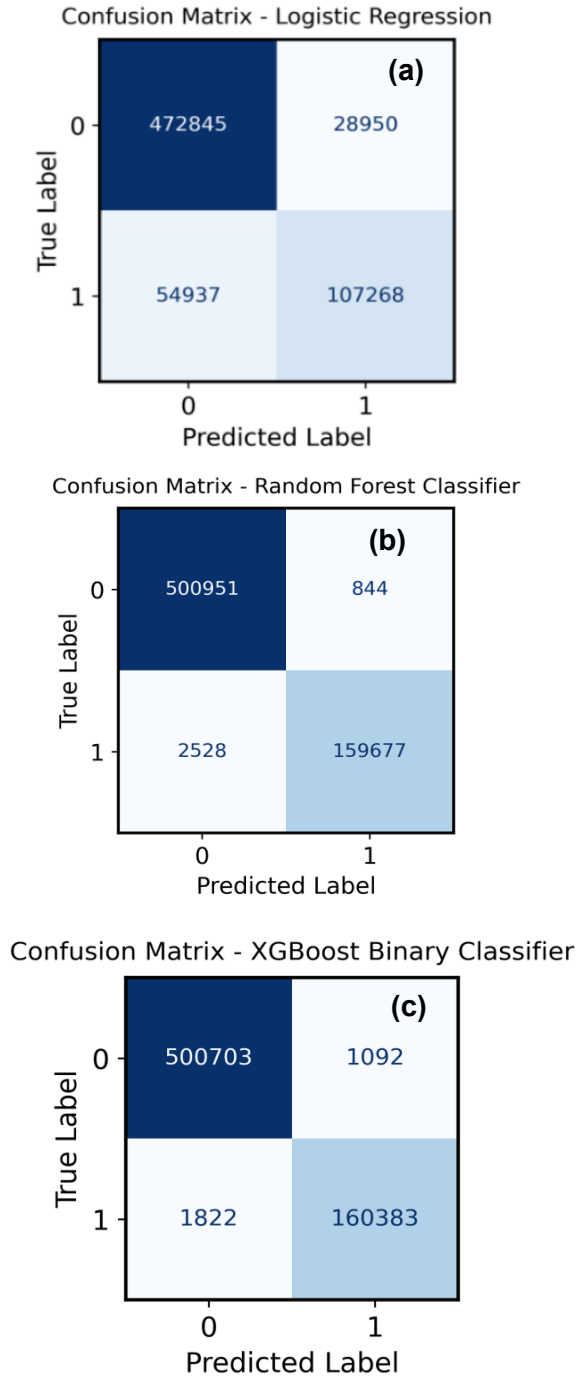


Fig.9. Confusion Matrices based on (a) LR, (b) RFC, and (c) XGBoost for binary classification.

TABLE III
COMPARISON OF KEY PERFORMANCE PARAMETERS AMONG PREVIOUS REPORTED WORKS AND THE PRESENT STUDY

Work	Method	Feature selection	Accuracy	Precision	Recall	F1 -score
2020 [Ref.32] Sahoo et al.	SVM+GA	KPCA	Dataset I – 98.71 Dataset II - 92.9	-	-	-
2020 [Ref.33] Tuan et al.	KNN	-	99.1	99.1	99.1	99.1
2020 [Ref.34] Liu et al.	CNN-LSTM	-	-	96.6	98.4	97.6
2023 [Ref.35] Songma et al.	XGBoost (Min-max normalization)	PCA	99.7	92.0	98.8	94.9
2024 [Ref. 36] Amali et al.	Decision Tree	EDA	98.5	-	-	-
2025 [Ref. 37] Cheekati et al.	SDPN	QRIME	93.3	91.4	98.5	94.8
2026 Present study	XGBoost	SelectKBest	99.6	99.3	98.9	99.1

V. CONCLUSION

This research puts forward experimental results obtained after training multiple state of the art ML models on a rich and varied DDoS dataset. The huge heterogeneity in current DDoS attack types is highlighted in this work and the need for a rich dataset encapsulating majority of the popular DDoS variants' characteristics is justified. Using powerful feature extraction techniques such as SelectKBest and extraction metrics such as mutual_info_classif, attempts have been made to obtain the most prominent features from the huge dataset of 80 columns. This dimension reduction is aimed at reducing noise and improving training and run times of the models on our prepared dataset. After a series of data pre-processing tasks and further attempts at refining, three ML models are trained on the finalised dataset. The models LR, RFC and XGBoost are chosen due to their ability to give optimal predictions on tabular datasets populated by numerical values. The obtained accuracy results lead to the conclusion that LR's metrics are too simplistic to properly identify the complex relations between network flow features of benign and malicious traffic, thus leading to a poor 87% accuracy. The RFC provides an impressive accuracy of 99.29% with a high AUC score - making it a suitable model for DDoS entry detection. The overall best accuracy of 99.56% was provided by the XGBoost model, which utilizes gradient boosting formulas for creating deeper connections between attribute values. The XGBoost model has a marginal upper hand over the RFC on overall accuracy, but it reduces false negatives by almost 33%, which is an important factor when it comes to comparing IDS models. The SHAPely values of the best performing model are chosen to see the impact of our chosen features and the result validates the SelectKBest's output. Results for other key performance metrics such as precision, recall and F1-score are obtained to be 99.30%, 98.92% and 99.11%, respectively. These results demonstrate high efficacy indication strong performance of our trained model. Thus, it can be concluded that current ML models such as XGBoost are effective in analysing attributes of recent and popular DDoS attack network properties and predicting whether a network flow is malicious or benign.

The present AI-driven model, which is capable of automatic recognition of the irregular traffic patterns with high accuracy, has significant practical impact in preventing network downtime, maintaining operational continuity, minimizing disruption and financial losses in business. This allows legitimate customers to access websites and applications. Mapping out the attack vectors gives the intelligence required to protect the systems against future and more complex attacks.

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